

COURSE SYLLABUS

CSC14115 – Applied Data Science

1. GENERAL INFORMATION

Course name:	Applied Data Science
Course name (in Vietnamese):	Khoa học dữ liệu ứng dụng
Course ID:	CSC14115
Knowledge block:	Specialized knowledge
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Data Science
Prior-course:	None
Instructors:	Le Ngoc Thanh

2. COURSE DESCRIPTION

This course is a continuation course in Data Science. Data Science course provides students with knowledge of Python programming, data science tools and processes with relatively simple methods. This course explores the case studies of the application of data science in reality. Thereby, students will see a wide range of practical applications of data science, learning more methods of data collection, pre-processing, and data analysis (besides methods in Data Science course). Students will be able to apply these methods to create a new application by themselves. In addition, the course will provide the necessary ethical principles when developing data science applications.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
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G1	Present and install case studies on application of data science	
G2	Apply data collection methods, apply data pre-processing methods	
G3	Apply data processing methods and evaluate the results	
G4	Propose suggestions for improvement	
G5	Apply presentation skills to communicate results clearly and effectively	
G6	Be aware of the importance of ethical issues when building data science applications	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Present case studies on application of data science	TU
G1.2	Install case studies on application of data science	TU
G2.1	Apply data collection methods	U
G2.2	Apply data pre-processing methods	U
G3.1	Apply data processing methods	U
G3.2	Evaluate the results	U
G4.1	Propose suggestions for improvement	TU
G5.1	Apply presentation skills to communicate results clearly and effectively	U

G6.1	Be aware of the importance of ethical issues when building data science applications	U
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5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Review and discussion Topics	G1.1	Lecturing
2	Data collection and visualization discussion	G2.1	Lecturing Demonstration, Q&A
3	Data preprocessing discussion	G2.2	Lecturing Demonstration, discussion
4	Machine learning discussion	G3.1	Lecturing Demonstration, discussion
5	Machine learning discussion (cont.)	G3.1	Lecturing Demonstration
6	Mid-Term Presentation	G1.1, G2.2, G6.1, G2.1, G5.1,	Question & answer Case study and discussion
7	Evaluation discussion	G3.2	Lecturing Demonstration
8	Libraries discussion	G3.2	Lecturing Q&A, discussion
9	Challenges and opportunities of DS	G4.1	Case study, discussion Demonstration

10	Challenges and opportunities of DS (cont.)	G4.1	Case study, discussion Demonstration
11	Final Project Presentation	G1.2, G3.2, G4.1, G5.1, G6.1	Discussion Project Presentation

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
A12	Homework: HW1-HW	HW: reading comprehension and writing reports in English	G2.1, G2.2, G3.1, G3.2, G4.1	20%
A13	Weekly discussion: DS1–DS10	Discussion in class	G2.1, G2.2, G3.1, G3.2, G4.1	10%
A2	Projects			70%
A21	MidTerm Project	Present about topic selection and data preparation	G1.1, G2.1, G2.2, G5.1, G6.1	20%

A22	Final Project	Present about topic results	G1.2, G3.2, G4.1, G5.1, G6.1	50%
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7. RESOURCES

Textbooks

No.	Textbook	Author	Pub. year	Genres	Link download/ Website
1	Machine Learning Foundations: A Case Study Approach	Carlos Guestrin, Emily Fox	2015	References	https://www.coursera.org/learn/ml-foundations
2	The Analytics Edge	Dimitris Bertsimas, Allison O'Hair	2016	References	https://www.edx.org/course/analytics-edge-mitx-15-071x-2
3	CS109 Data Science	Joe Blitzstein, Hanspeter Pfister, Verena Kaynig-Fittkau	2015	References	http://cs109.github.io/2015/
4	Data Science Ethics	H.V. Jagadish	2016	References	https://www.coursera.org/learn/data-science-ethics

Others

- Slides

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.